LECTURE 1.5 · SHIP AND OCEAN SYSTEMS DESIGN

The Design Equation

Papanikolaou (2026) §2.13, Appendix C · Session W1.4 · Lecture 20 min + Exercise 1.5 60 min

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One Equation, Two Sides

Buoyancy: the hull

$$\Delta = \rho\, C_B\, L\, B\, T\, k_A$$

kA corrects the moulded volume for shell plating and appendages.

Weight: the ship and its load

$$\Delta = \mathrm{DWT} + W_{ST} + W_{OT} + W_M$$

Steel, outfit and machinery all grow with the ship, so Δ sits on both sides.

In Exercise 1.4, Δ came from the deadweight and L, B, T from separate regressions. Put those L, B and T in the buoyancy side and you do not get back the same Δ: the two sides **miss by a few percent**. Today we choose the size so that both sides give the same Δ: that is the design equation.

Fix the Ratios, and One Dimension Is Left

Deadweight carriers: from Δ

Take L/B and B/T from similar ships and write L and T through B:

$$L = \left(\frac{L}{B}\right) B, \qquad T = \frac{B}{(B/T)}$$

Buoyancy then has one unknown, B:

$$\Delta = \rho\, C_B\, k_A\, \frac{(L/B)}{(B/T)}\, B^3$$

$$B = \left[\frac{\Delta\,(B/T)}{\rho\, C_B\,(L/B)\, k_A}\right]^{1/3} \qquad L = \left[\frac{\Delta\,(L/B)\,(L/T)}{\rho\, C_B\, k_A}\right]^{1/3}$$

Then T = B/(B/T) and D = B/(B/D).

Volume carriers: from the hull volume

The same steps, with B/D in place of B/T:

$$B = \left[\frac{\nabla_D\,(B/D)}{C_{BD}\,(L/B)}\right]^{1/3}$$

Note · the hull volume to the deck

$$\nabla_D = C_{BD}\, L\, B\, D$$

Moulded volume up to the main deck, at depth D: not the displaced volume, which stops at the draught T.

$$C_{BD} = C_B + C_1\, \frac{D - T}{T}\,(1 - C_B)$$

$C_1 \approx 0.25$ for little flare above the waterline, 0.4 to 0.7 for strong flare.

The Reference Ship: Closing the Gap

DWT 37,600 t, 14.5 kn, draught limit 11.5 m (0.6, slide 2) · tanker fits of Appendix A.4 · ρ = 1.025 t/m³, kA = 1

Given by Exercise 1.4 (partially solved in class)

Δ = 46,050 t (Fig. A.51) · L 174.7 · B 29.50 · T 10.97 · D 14.95 m (Figs. A.53–A.56)

L/B 5.92 · B/T 2.69 · B/D 1.97 · CB 0.757 (Horn, Fn 0.180)

1	The gap: buoyancy with those values	$\rho C_B L B T = 1.025 \times 0.757 \times 174.7 \times 29.50 \times 10.97 = 43{,}880\ \text{t} \quad (-4.7\,\%)$
2	Solve the design equation for B	$B = \left[\dfrac{46{,}050 \times 2.69}{1.025 \times 0.757 \times 5.92} \right]^{1/3} = 30.0\ \text{m}$
3	L, T and D from the ratios	$L = 5.92 \times 30.0 = 177.6\ \text{m}, \quad T = \dfrac{30.0}{2.69} = 11.15\ \text{m}, \quad D = \dfrac{30.0}{1.97} = 15.2\ \text{m}$
4	Check: both sides and the brief	$\rho C_B L B T = 46{,}050\ \text{t} = \Delta, \qquad T = 11.15\ \text{m} < 11.5\ \text{m}$

Normand: Each Weight Group Scales with Δ

Weight group	Grows with	Power of Δ
Steel structure, outfitting	the hull	1
Machinery	$P \propto \Delta^{2/3} \cdot V^3$	2/3
Fuel	$P \cdot R/V$	2/3
Crew and effects	installed power	2/3
Payload, passengers, stores	the owner	0

$$\Delta = A\,\Delta + B\,\Delta^{2/3} + C$$

A, B and C come from a parent ship whose weight groups are known.

Valid for deviations from the parent of up to **10%**, exceptionally 20%. Beyond that, exponents by ship type: steel 0.92 for tankers, 0.98 for bulk carriers.

Solving It: Iterate Until the Ship Floats

1

Write it as

$$\Delta = g(\Delta)$$

$$g(\Delta) = \mathrm{DWT} + W_L(\Delta)$$

The weight of a ship of size Δ. Start at Δ_0 = DWT, a ship with no lightship.

2

Iterate

$$\Delta_{k+1} = g(\Delta_k)$$

Add up the weights of the current ship; that total is the displacement of the next one.

3

Check

$$|g'(\Delta)| < 1$$

Each extra tonne of ship must cost less than a tonne of its own lightship, or the loop never closes.

$$g'(\Delta) = \frac{1}{\Delta} \sum_i n_i \, W_i$$

n_i is the exponent of weight group W_i. With exponents of 1 or less and a lightship smaller than Δ, g′ stays below 1: the iteration converges. Newton–Raphson converges faster: homework H1.

Review of numerical methods

Theory · Fixed-point iteration (successive substitution, Picard)

$f(x) = 0 \iff x = g(x) \qquad x_{k+1} = g(x_k)$

$|g'(x)| < 1 \text{ near } x^* \;\Rightarrow\; e_{k+1} \approx g'(x^*)\, e_k$

Needs only g. Error × |g′(x*)| per pass: **linear**.

Theory · Newton–Raphson (tangent method)

$f(x) = 0 \qquad x_{k+1} = x_k - \dfrac{f(x_k)}{f'(x_k)} \qquad f'(x^*) \neq 0 \;\Rightarrow\; e_{k+1} \approx \dfrac{f''(x^*)}{2f'(x^*)}\, e_k^2$

Needs f and f′, a start close enough. Error squared: **quadratic**.

Example $\Delta = \underbrace{10\,000}_{\text{DWT}} + \underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · stop when the change is **below 0.01 t**

To compose the fixed-point iteration

1 · f = 0 $f(\Delta) = \Delta - 10\,000 - 150\sqrt{\Delta} = 0 \iff \Delta = 10\,000 + 150\sqrt{\Delta}$

2 · g, g′ $g(\Delta) = 10\,000 + 150\sqrt{\Delta} \quad g'(\Delta) = \dfrac{75}{\sqrt{\Delta}} \le 0.75 < 1$ ✓

3 · iterate $\Delta_{k+1} = 10\,000 + 150\sqrt{\Delta_k}$

To compose Newton–Raphson

1 · f, f′ $f(\Delta) = \Delta - 10\,000 - 150\sqrt{\Delta} \quad f'(\Delta) = 1 - \dfrac{75}{\sqrt{\Delta}} \ge 0.25 \neq 0$ ✓

2 · iterate $\Delta_{k+1} = \Delta_k - \dfrac{\Delta_k - 10\,000 - 150\sqrt{\Delta_k}}{1 - 75/\sqrt{\Delta_k}}$

Same root, 40 000 t: 17 evaluations of g against 5 Newton steps

Example $\Delta = \underbrace{10\,000}_{\text{DWT}} + \underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · count the evaluations of g until the change is **below 0.01 t**

k	Δ_k	$150\sqrt{\Delta_k}$	Δ_{k+1}	change	ratio
0	10 000.0	15 000.0	25 000.0	+15 000.0	
1	25 000.0	23 717.1	33 717.1	+8 717.1	0.581
2	33 717.1	27 543.3	37 543.3	+3 826.2	0.439
3	37 543.3	29 064.1	39 064.1	+1 520.8	0.397
⋮	⋮	⋮	⋮	⋮	⋮
14	39 999.95	29 999.98	39 999.98	+0.032	0.375
15	39 999.98	29 999.99	39 999.99	+0.012	0.375
16	39 999.99	30 000.00	40 000.00	+0.005 ✓	0.375

Fixed-point iteration: 17 evaluations of g (k = 0 to 16). Change × 0.375 = g′(Δ*) per pass: linear.

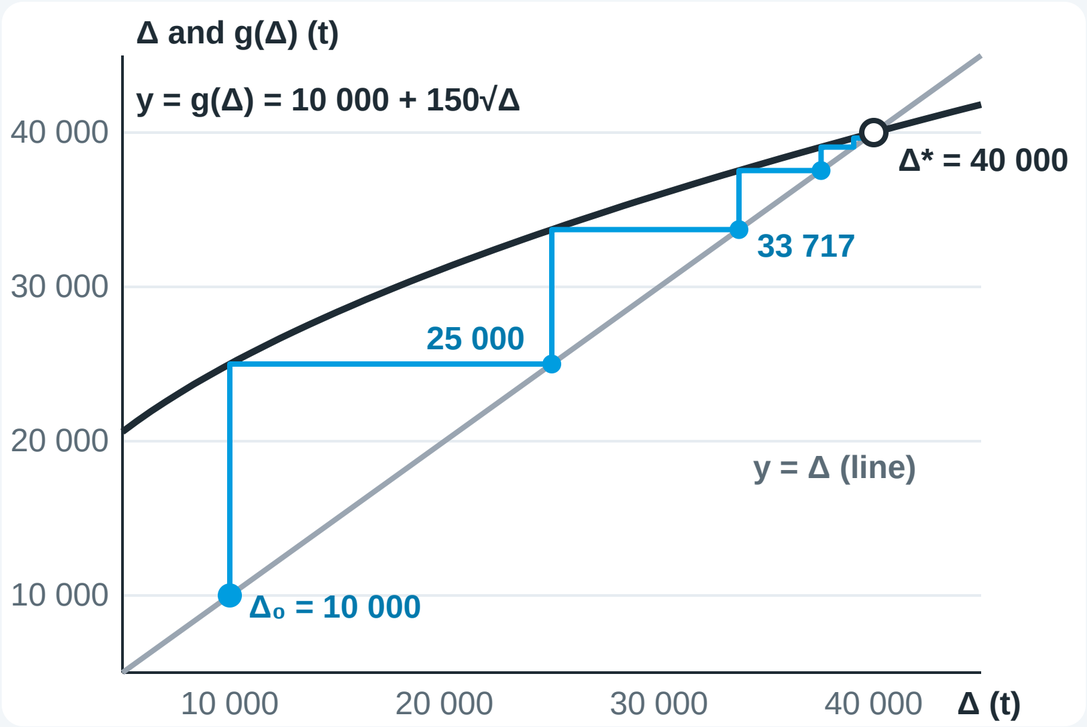
k	Δ_k	$f(\Delta_k)$	$f'(\Delta_k)$	Δ_{k+1}	change
0	10 000.0	−15 000.0	0.250	70 000.0	+60 000.0
1	70 000.0	+20 313.7	0.717	41 649.7	−28 350.3
2	41 649.7	+1 037.3	0.633	40 009.7	−1 640.0
3	40 009.7	+6.05	0.625	40 000.0	−9.68
4	40 000.0	+0.0002	0.625	40 000.0	−0.0004 ✓

k = 0: 15 000 t short, slope only 0.25: grow by 15 000 / 0.25 = 60 000 t, far past Δ*.

Newton–Raphson: 5 steps (k = 0 to 4), each one evaluation of g and g′. Once close, error 1 650 → 9.7 → 0.0004 t: squared each step, quadratic.

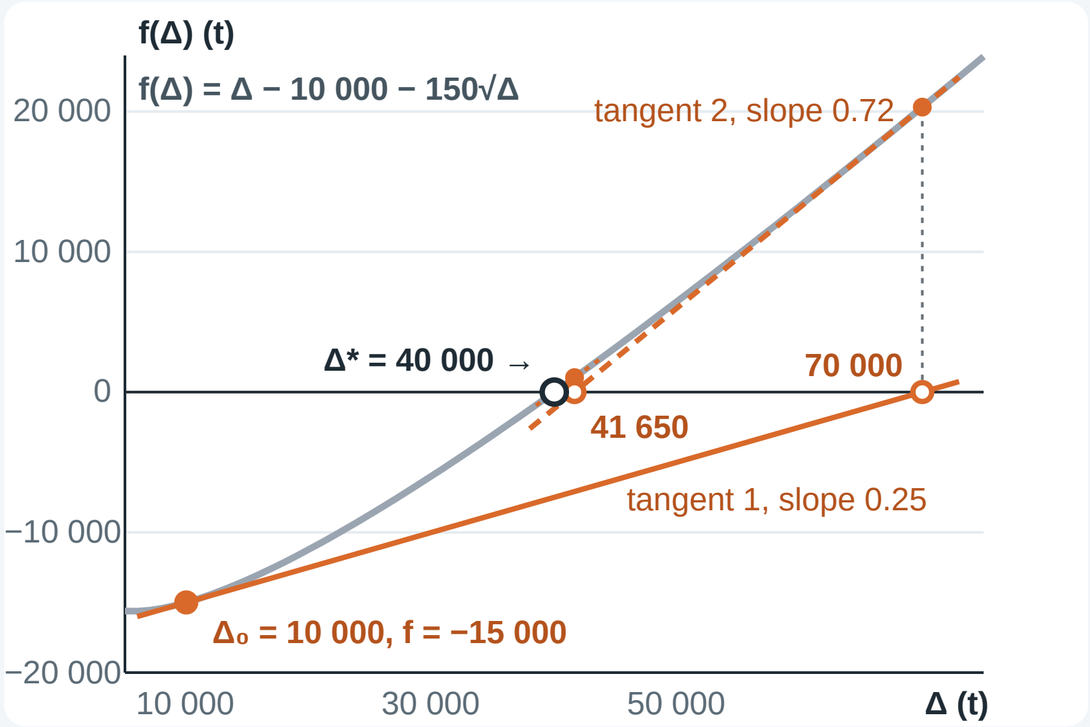
Fixed point climbs a staircase; Newton follows tangents

Fixed-point iteration · Δ = g(Δ)



Δ = g(Δ) = DWT + WLS(Δ): the equilibrium is “buoyancy = weight computed from the current size”.

Newton–Raphson · f(Δ) = 0



f(Δ) = buoyancy − weight: the equilibrium is “excess buoyancy = 0”.

Sizing the reference tanker: 37 600 t deadweight

Objective: determine the displacement Δ and the main dimensions using simplified model equations and an iterative solution. The owner’s brief (0.6) fixes the deadweight and the service speed; the proportions and the block coefficient come from Exercise 1.4. The weight of the ship is assumed to break down as:

$$W = \mathrm{DWT} + \underbrace{W_{\mathrm{st}} + W_{\mathrm{o}} + W_{\mathrm{m}}}_{W_{\mathrm{LS}}}$$

Steel	$W_{\mathrm{st}} = K\,E^{1.36} \quad E = L\,(B + T) + 0.85\,L\,(D - T)$
Outfit	$W_{\mathrm{o}} = c_{\mathrm{o}}\,\Delta$
Machinery	$W_{\mathrm{m}} = k_{\mathrm{m}}\,P_{\mathrm{B}} \quad P_{\mathrm{B}} = \dfrac{\Delta^{2/3}\,V^{3}}{C_{\mathrm{adm}}}$

Item	Symbol	Value
Deadweight	DWT	37 600 t
Service speed	V	14.5 kn
Seawater density	ρ	1.025 t/m³
Block coefficient (Ex 1.4)	C_{B}	0.757
Length / beam (Ex 1.4)	L/B	5.92
Beam / draught (Ex 1.4)	B/T	2.69
Depth / draught (Ex 1.4)	D/T	1.36
Steel coefficient (tankers)	K	0.032
Outfit coefficient	c_{o}	0.025
Machinery weight	k_{m}	0.10 t/kW
Admiralty coefficient	C_{adm}	450

EXERCISE 1.5 · QUESTIONS 1–2

Buoyancy = weight; Δ sets the dimensions

Q1 · Equilibrium equation

Write displacement from buoyancy computation = displacement from weight components.

$$\rho\, C_{\mathrm{B}}\, L\, B\, T = \mathrm{DWT} + W_{\mathrm{st}} + W_{\mathrm{o}} + W_{\mathrm{m}}$$

- Known: DWT = 37 600 t. Unknown: Δ.
- Lightship > 0, so Δ > DWT.
- Δ_0 = DWT (zero lightship) is a lower bound and a natural start.

Q2 · Displacement equation as function of non-dimensional ratios

Write Δ with L/B, B/T and the block coefficient; then T(Δ).

1 · Inject the ratios (in blue) into the displacement equation

$$B = (B/T)\,T \qquad L = (L/B)\,B = (L/B)\,(B/T)\,T$$

$$\Delta = \rho\, C_{\mathrm{B}}\, \underbrace{(L/B)\,(B/T)\,T}_{L}\, \underbrace{(B/T)\,T}_{B}\, T = \rho\, C_{\mathrm{B}}\, (L/B)\,(B/T)^2\, T^3$$

2 · Draught from Δ, then B, L and D = (D/T)·T

$$T = \left(\frac{\Delta}{\rho\, C_B\, (L/B)\,(B/T)^2} \right)^{1/3} = 0.3110\, \Delta^{1/3}$$

For a given Δ, with L/B and B/T fixed, L, B and T are determined.

3 · Similar equations follow with L or B as the variable

$$\Delta = \frac{\rho\, C_B\, L^3}{(L/B)^2\,(B/T)} \;\Rightarrow\; L = 4.953\, \Delta^{1/3} \qquad \Delta = \frac{\rho\, C_B\, (L/B)\, B^3}{(B/T)} \;\Rightarrow\; B = 0.8366\, \Delta^{1/3}$$

Everything becomes a power of Δ

Q3 · Steel weight

Write the steel weight as a power of Δ and find its coefficients.

$$W_{\text{st}} = K\,E^{1.36} \qquad E = L\,(B+T) + 0.85\,L\,(D-T)$$

$$E = L^2\left[\frac{1}{(L/B)} + \frac{1}{(L/B)(B/T)} + \frac{0.85\,\big((D/T)-1\big)}{(L/B)(B/T)}\right] = 0.2509\,L^2$$

$$E = 0.2509\,(4.953)^2\,\Delta^{2/3} = c_E\,\Delta^{2/3}$$

$$W_{\text{st}} = K\,c_{\text{E}}^{1.36}\,\Delta^{\frac{2}{3}\times 1.36} = c_{\text{st}}\,\Delta^{n}$$

$$c_E = 6.156, \qquad c_{st} = 0.3789, \qquad n = 0.907$$

Q4 · Iteration function

Write g(Δ) with all coefficients evaluated, for fixed-point iteration

Equilibrium with the buoyancy in non-dimensional form and every weight as a power of Δ:

$$\underbrace{\rho\,C_{\text{B}}\,(L/B)\,(B/T)^2\,T^3}_{\Delta} = \text{DWT} + c_{\text{o}}\,\Delta + c_{\text{st}}\,\Delta^{n} + b\,\Delta^{2/3}$$

$$b = \frac{k_m\,V^3}{C_{adm}} = 0.6775$$

$$\Delta_{k+1} = g(\Delta_k) = \text{DWT} + c_{\text{o}}\,\Delta_k + c_{\text{st}}\,\Delta_k^{n} + b\,\Delta_k^{2/3}$$

Q5 · Without iterating, prove that the iteration converges for any Δ ≥ DWT, and interpret g′(Δ).

Why it converges: g′ stays below 1

Criterion	$x_{k+1} = g(x_k)$ converges to x^* if $\lvert g'(x)\rvert < 1$ in a region containing x^*
Our problem	$x = \Delta, \quad \Delta^* > \mathrm{DWT} \Rightarrow$ show $\lvert g'(\Delta)\rvert < 1$ for all $\Delta \geq \mathrm{DWT}$

$$\Delta_{k+1} = g(\Delta_k) = \mathrm{DWT} + c_o\,\Delta_k + c_{st}\,\Delta_k^{\,n} + b\,\Delta_k^{\,2/3}$$

Differentiate term by term :

$$g'(\Delta) = c_o + n\,c_{st}\,\Delta^{n-1} + \tfrac{2}{3}\,b\,\Delta^{-1/3} = 0.025 + 0.3436\,\Delta^{-0.093} + 0.4516\,\Delta^{-1/3}$$

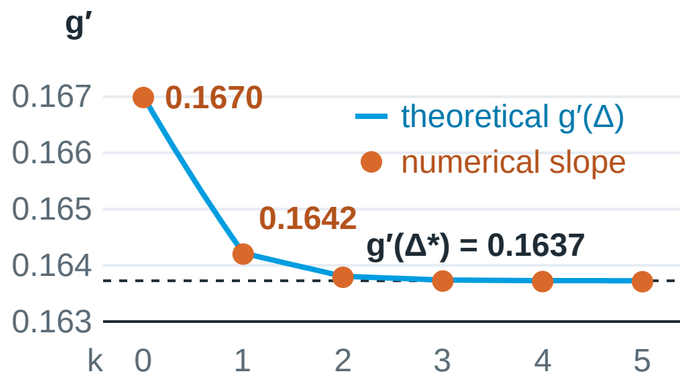
All terms are positive and decrease with Δ, so the maximum is at Δ = DWT:

$$g'(\mathrm{DWT}) = 0.025 + 0.3436 \times 37\,600^{-0.093} + 0.4516 \times 37\,600^{-1/3}$$
$$g'(\Delta) \leq g'(\mathrm{DWT}) = 0.025 + 0.1285 + 0.0135 = 0.167$$

Physical meaning: each extra tonne of displacement costs only 0.17 t of its own lightship, leaving **0.83 t** to float the shortfall.

With g′ > 1, every enlargement would sink the ship deeper.

k	Δ_k	$g(\Delta_k)$	$g(\Delta_k+100)$	slope
0	37 600.0	44 630.3	44 647.0	**0.1670**
1	44 630.3	45 794.3	45 810.7	**0.1642**
2	45 794.3	45 985.2	46 001.6	**0.1638**
3	45 985.2	46 016.5	46 032.9	**0.1637**
4	46 016.5	46 021.6	46 038.0	**0.1637**
5	46 021.6	46 022.4	46 038.8	**0.1637**



Q6 · Iterate g(Δ) by hand from Δ_0 = DWT until the change is below 1 t. Give Δ*, dimensions and weights.

46 022.6 t in six evaluations of g

$$\Delta_{k+1} = \underbrace{37\,600}_{\text{DWT}} + \underbrace{0.3789\,\Delta_k^{0.907}}_{\text{steel}} + \underbrace{0.025\,\Delta_k}_{\text{outfit}} + \underbrace{0.6775\,\Delta_k^{2/3}}_{\text{machinery}}$$

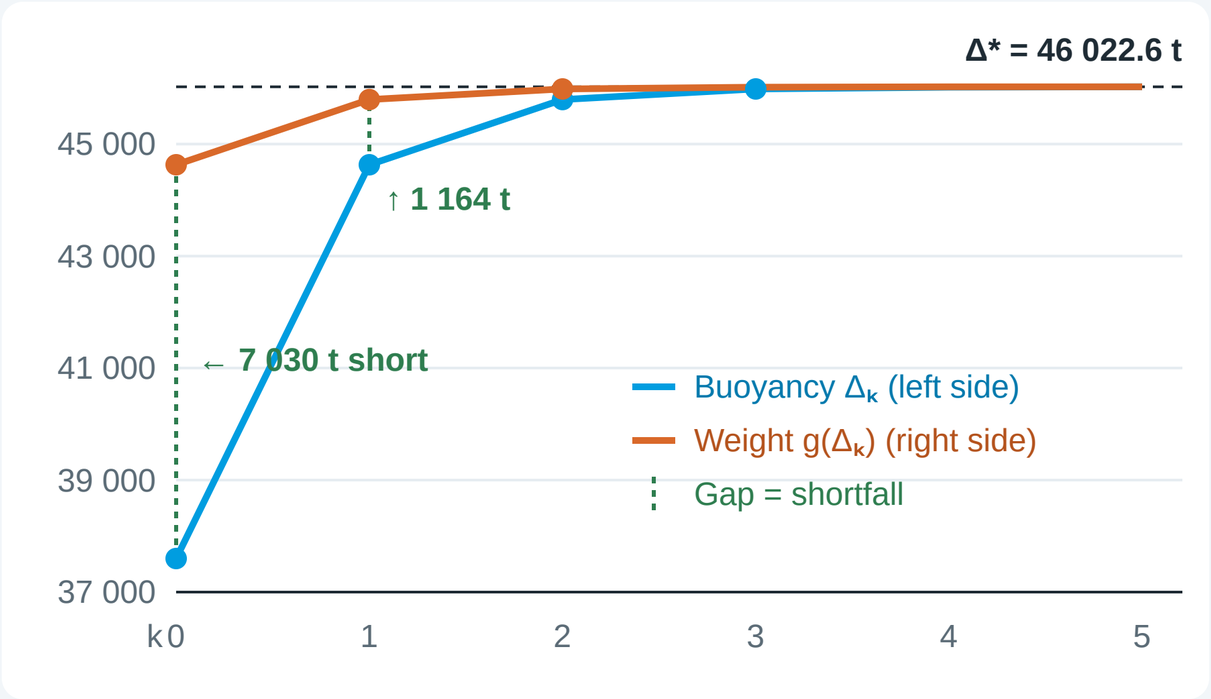
Coefficients used: L/B = 5.92 · B/T = 2.69 · D/T = 1.36 · block coefficient = 0.757 · ρ = 1.025 t/m³

k	Δ_k (t)	T (m)	L (m)	E (m²)	Steel	Outfit	Machinery	Lightship	Δ_{k+1} (t)	Change (t)
	given	0.3110·Δ^⅓	(L/B)(B/T)·T	0.2509·L²	0.032·E^1.36	0.025·Δ	0.6775·Δ^⅔	Σ weights	DWT + Σ	$\Delta_{k+1} - \Delta_k$
0	37 600.0	10.420	165.93	6 909	5 330.0	940.0	760.4	**7 030.3**	44 630.3	7 030.3
1	44 630.3	11.032	175.69	7 745	6 226.2	1 115.8	852.4	**8 194.3**	45 794.3	1 164.0
2	45 794.3	11.127	177.20	7 879	6 373.2	1 144.9	867.2	**8 385.2**	45 985.2	190.9
3	45 985.2	11.143	177.45	7 901	6 397.3	1 149.6	869.6	**8 416.5**	46 016.5	31.3
4	46 016.5	11.145	177.49	7 905	6 401.2	1 150.4	870.0	**8 421.6**	46 021.6	5.1
5	46 021.6	11.146	177.49	7 905	6 401.9	1 150.5	870.0	**8 422.4**	46 022.4	0.8 < 1 ✓

k = 0 → 5: Δ grows +8 422 t but the lightship only +1 392 t (ratio 0.165 ≈ g′) → next slide. Weights in t.

Q6 · continued: both sides of the equation, step by step

The lightship lags behind: +0.16 t per tonne



Step	Δ grows	Lightship grows	Ratio
0 → 1	+7 030.3	**+1 164.0**	**0.166**
1 → 2	+1 164.0	**+190.9**	**0.164**
2 → 3	+190.9	**+31.3**	**0.164**
3 → 4	+31.3	**+5.1**	**0.164**
4 → 5	+5.1	**+0.8**	**0.164**
0 → 5	**+8 421.6**	**+1 392.1**	**0.165**

$$\text{ratio} = \frac{LS(\Delta_k) - LS(\Delta_{k-1})}{\Delta_k - \Delta_{k-1}} \;\to\; g'(\Delta^*) = 0.164$$

The ratio is the **average slope of g over one step**, equal to g′ at a point inside it. Question 5 proved g′ ≤ 0.167, so each shortfall is at most 17% of the previous one. Weights in t.

Three Things to Take Away

The design equation turns buoyancy = weight into one dimension, once the ratios and CB come from similar ships.

Normand writes every weight group as a power of Δ, from a parent ship within 10%:

$$\Delta = A\,\Delta + B\,\Delta^{2/3} + C$$

Iterate from Δ_0 = DWT; it converges because the lightship grows more slowly than the ship.

Next: Wed 11 Nov, human factors in ship design (guest lecturer). Fri 13 Nov (W2.3): length, beam, depth, draught and form coefficients. Read §2.3–2.12.